Anchor Representation: 2024 vs. 2026

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Overview

This report compares anchor representation between the 2024 and 2026 calibration datasets.

The purpose is to determine which anchors were sufficiently represented in each calibration period and which anchors were excluded because they did not meet the representation criterion.

For both 2024 and 2026, an anchor is considered sufficiently represented if it appears in at least 80% of the available calibration files.

This analysis distinguishes between:

- anchors retained in both analyses
- anchors retained in 2024 but not 2026
- anchors retained in 2026 but not 2024
- anchors that appeared in the raw data but were excluded from both analyses

1. 2024 Calibration Files

1.1 Load 2024 Files

- Total number of available 2024 calibration files: 29

1.2 Read 2024 Anchor IDs

- IDs are immediately converted to char to be consistent with 2026 data

1.3 Apply the 80% Representation Criterion to 2024 Data

- ID must appear in at least 80% of the available 2024 calibration files: 23.2 of 29 files
- Number of retained 2024 anchors: 672

2. 2026 Calibration Files

2.1 Load 2026 Files

- Total number of available 2026 calibration files: 89

We impose a filtering technique to ensure robustness for our comparatively wider period of calibration data collection for 2026. A number of datasets of the 2026 calibration data contain significantly few IDs, and would inhibit our analysis by presenting many missing data points. The 2026 representation analysis therefore uses only files containing at least 500 IDs.

- Number of 2026 files retained: 80

2.2 Read 2026 Anchor IDs

2.3 Apply the 80% Representation Criterion to 2026 Data

- ID must appear in at least 80% of the usable 2026 calibration files: 64 of 80 files
- Number of retained 2026 anchors: 899

3. Anchor Retention

3.1 Anchors Retained in Both Years

Number retained in both: 535

3.2 Anchors Retained in 2024 but Not 2026

137 anchors met the 80% criterion in 2024 but did not meet it in 2026.

3.3 Anchors Retained in 2026 but Not 2024

364 met the 80% criterion in 2026 but did not meet it in 2024.

3.4 All Raw Anchors Observed in 2024

691 unique anchors appeared at least once in the original 2024 calibration files.

3.5 All Raw Anchors Observed in 2026

965 unique anchors appeared at least once in the usable 2026 calibration files.

3.6 Anchors Excluded from Both Analyses

An anchor is considered excluded from both analyses if:

- it appeared in the raw 2024 and/or 2026 data
- it did not meet the 80% criterion in 2024
- it did not meet the 80% criterion in 2026

Unique anchors excluded from both: 65

4. Anchor Representation Summaries

4.2 Anchor Retention Results

category	number_of_anchors	% of all observed
Retained in both 2024 and 2026	535	0.4859219
Retained in 2024 only	137	0.1244323
Retained in 2026 only	364	0.3306085
Excluded from both; present in both raw datasets	0	0.0000000
Excluded from both; present in 2024 only	12	0.0108992
Excluded from both; present in 2026 only	53	0.0481381

4.2 Most Poorly Represented Anchors

The following identifies anchors with the lowest representation in either year; only the first 6 of 65 rows are printed.

id	representation_2024	representation_2026	minimum_representation	classification
6025	1.0000000	0	0	Retained in 2024 only
6039	1.0000000	0	0	Retained in 2024 only
6042	0.7586207	0	0	Excluded from both; 2024 only
6043	1.0000000	0	0	Retained in 2024 only
6054	1.0000000	0	0	Retained in 2024 only
6072	1.0000000	0	0	Retained in 2024 only

4.3 Final Interpretation

1. **Stable/repeated anchors:** retained in both 2024 and 2026 analyses
2. **Year-specific anchors:** retained in one calibration period but not the other
3. **Intermittent anchors:** observed in the raw data but excluded entirely

The "Intermittent anchors" group represents anchors with consistent changes in anchor availability, connectivity, or geographic coverage. In order to reduce computational overhead and accuracy, a first step may be to exclude these anchors as input to our geolocation algorithm.

4.4 CSVs of Results